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A simple and fast method to calculate the coil currents and the external poloidal flux and magnetic field for fixed boundary equilibria

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Abstract

A simple and fast method to calculate the coil currents and the external poloidal flux and magnetic field consistent with a given internal magnetohydrodynamic (MHD) equilibrium and coil geometry is presented. The constrained optimization technique employed allows to fix the exact position of the X-points. The external poloidal flux and field are calculated using Green's functions expressed in terms of elliptic integrals that are calculated using a fast and accurate routine. The calculation can be limited to the region between the plasma and the device wall. An example for a double null configuration is presented.

Keywords: MHD equilibrium, Green's functions, tokamak

(Some figures may appear in colour only in the online journal)

1. Introduction

A magnetic fusion experiment or a commercial reactor can be considered successful if it produces a plasma that is stable (not destroyed by macroscopic instabilities), remains confined for a long enough time at the desired temperature and density (it has low enough transport) and confines the high energy products produced by fusion reactions (i.e. alpha particles) until they transfer a large enough fraction of their energy to the plasma. Since the plasma reaches an ideal MHD equilibrium in a very short time, the study of these three important aspects, and many others, requires calculating first the equilibrium magnetic field and the density and temperature distribution inside the plasma and the magnetic field in the region surrounding the plasma. In other words, it is necessary to know first the internal plasma and magnetic

field equilibrium and the external magnetic field.

A complete equilibrium calculation requires calculating the magnetic field, poloidal flux and plasma profiles inside the closed flux region and the flux and field in the vacuum region surrounding the plasma (at least in the region between the last closed flux surface and the material wall). In principle, two different problems can be considered. In one case, the coil geometry, the internal magnetic field, the plasma profiles and the shape of the plasma boundary are fixed (fixed boundary) and it is necessary to calculate the currents in the external coils and the flux and field outside the plasma. In the other case, the coil geometry and the currents flowing in these coils are known but the position and shape of the plasma boundary are not known, so it is necessary to solve a free boundary problem. In general, the free boundary problem is solved iteratively, by alternating the calculation of the equilibrium inside and outside a plasma boundary which is updated after each iteration. A related problem is the reconstruction of the equilibrium from the experimen-

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tal information available.

The very large difference between the time it takes the plasma to reach an MHD equilibrium and that associated to transport processes has allowed the development of computer codes that can follow the evolution of the plasma during the discharge. These codes alternate between the calculation of the 2D MHD equilibrium and the solution of the 1D transport equations, obtained by averaging all quantities over the flux surfaces corresponding to the equilibrium calculated in the previous time step. Examples of such codes are DINA [1, 2] and TSC [3]. A different, faster, method is employed in the TOKSCEN [4] and CEDRES [5] codes, where the plasma current density profile is specified as a function of time and poloidal flux. These codes usually perform a “free boundary” calculation, where the currents in the external poloidal coils are specified and the plasma boundary is determined by the combined contribution of these currents and the plasma current (the eddy currents flowing in the passive elements are also calculated in some codes). More recently, integrated simulation environments have been developed that include numerical codes for calculating the equilibrium, stability and evolution of the plasma, plus a code for generating three-dimensional grids in regions of complex shape [6]. This package, named SIEMNED, includes a browser client and a server part that performs the calculations, stores the data and exchanges information between the different modules.

The calculation of the coil currents for a fixed plasma equilibrium appears to be a relatively simple problem. The poloidal flux at the plasma boundary is the sum of the flux produced by the current flowing in the plasma and the currents flowing in the external coils, $\psi_b = \psi_p + \psi_c$, where ψ_b is the boundary flux and ψ_p and ψ_c are the plasma and coils contributions respectively. If the boundary flux corresponding to the internal equilibrium is ψ_{in} the matching condition results $\psi_b = \psi_{in}$. Since the plasma contribution can be calculated from the known plasma current density and the contribution of each coil to the flux is proportional (linear) to the current that circulates in it, the matching condition can be used to generate a system of linear algebraic equations for the coil currents. If there are N coils, it would be enough to apply the condition at N points on the plasma boundary to obtain a system of N linear algebraic equations to calculate the coil currents [7].

Unfortunately, the simple procedure described above does not work in general because it leads to an ill posed problem [8]. The poloidal flux calculated with the coil currents obtained from the solution of the system of algebraic equation satisfies the boundary conditions imposed but may not be physically acceptable because it does not have the desired structure of closed flux surfaces in the plasma region. The method usually employed to calculate the coil currents consists of choosing a large number of points (≈ 30) along the plasma boundary and introducing an error function which is defined as the difference between the calculated flux (ψ_b) and the desired flux (ψ_{in}) [5, 9, 10]. The coil currents are then obtained by minimizing this error. A similar procedure is employed in the codes used to control the

circuits that drive the coil currents during the discharge. In this case, a set of probes measures the flux at various locations inside the chamber and the difference between the measured and desired values is used to adjust the coil currents [11]. Once the coil currents are known the external poloidal flux and magnetic field can be calculated using finite differences, finite elements or Green’s functions.

In this paper, we present a simple and fast method to calculate the coil currents and the external poloidal flux and magnetic field for a given internal MHD equilibrium and coil geometry. That is, we assume that the toroidal current density distribution inside the plasma, the shape and position of the plasma boundary (last closed flux surface, LCFS) and the positions of the coils are known and calculate the required coil currents and the resulting external flux and poloidal field. The method employed to calculate the internal equilibrium, analytical or numerical, is not relevant.

Our method involves a constrained optimization process and has the following advantages:

- (1) Both the external poloidal flux and the poloidal field are calculated in terms of the same elliptic integrals. It is not necessary to calculate derivatives of the poloidal flux to obtain the components of the poloidal field. This avoids possible numerical errors introduced in the calculation of the field and is very useful when calculating the trajectories of particles that escape from the plasma.
- (2) The use of Lagrange multipliers allows us to enforce boundary conditions for the flux and both components of the poloidal field at the X-points. This results in the X-points being at the exact prescribed position.
- (3) It is not necessary to calculate the external flux and field in a very large region, typically a semicircle much larger than the plasma, to enforce the boundary condition at infinity.
- (4) A very fast and accurate routine is employed to calculate the elliptic integrals [12].

The rest of this paper is organized as follows. Section 2 describes the method employed to calculate the currents in the external coils and the external flux and field. Section 3 presents an example, where the method is applied to a double null symmetric equilibrium and section 4, the conclusions.

2. Currents in the external coils and external flux and field

As already indicated, we assume that the internal MHD equilibrium, including the plasma boundary, is known and remains fixed. The problem we have to solve is as follows. Given the positions of a set of external poloidal field coils calculate the currents that have to flow in these coils to produce a total poloidal flux at the boundary of the plasma, sum of the contributions of the external coils and the current flowing inside the plasma, that is as close as possible to the boundary value corresponding to the internal equilibrium. Once the coil currents are known, the total field and flux in the vacuum regions can be calculated. The standard proce-

ture to calculate the coil currents consists of introducing a function that measures the difference between the desired and calculated boundary values and making it as small as possible.

There are two contributions to the boundary flux, one comes from the current flowing inside the plasma ψ_p and the other from the currents in the external coils ψ_c .

$$\psi_b(R_b, Z_b) = \psi_p(R_b, Z_b) + \psi_c(R_b, Z_b), \quad (1)$$

where cylindrical coordinates (R, ϕ, Z) will be employed and (R_b, Z_b) are the coordinates of a point on the plasma boundary.

In a static axisymmetric MHD equilibrium the pressure and the poloidal current are functions of the poloidal flux. In addition, the equilibrium condition relates the poloidal flux with the toroidal current density J_ϕ inside the plasma. Introducing the major radius (R_0) and the vacuum toroidal field at R_0 (B_0) and normalizing all lengths with R_0 , the poloidal flux with $2\pi B_0 R_0^2$, the pressure (p) with B_0^2/μ_0 , the poloidal current function (F) with $B_0 R_0$ and the toroidal current density with $B_0/\mu_0 R_0$ we obtain the following Grad-Shafranov equation [13],

$$\frac{\partial^2 \psi}{\partial R^2} - \frac{1}{R} \frac{\partial \psi}{\partial R} + \frac{\partial^2 \psi}{\partial Z^2} = -R^2 \frac{dp}{d\psi} - F \frac{dF}{d\psi} = -RJ_\phi, \quad (2)$$

where all quantities are dimensionless. Different equilibria are obtained for different choices of $p(\psi)$ and $F(\psi)$. The components of the poloidal field are related to the derivatives of the poloidal flux, $B_R = -(1/R)\partial\psi/\partial Z$, $B_Z = (1/R)\partial\psi/\partial R$. The contribution of the plasma current to the poloidal flux at the boundary is calculated using the corresponding Green's function.

$$\psi_p(R_b, Z_b) = \int_{\Omega_p} J_\phi(R, Z) G^\psi(R, Z, R_b, Z_b) dR dZ, \quad (3)$$

where $G^\psi(R, Z, R_b, Z_b)$ is the Green's function for the poloidal flux and the integration extends over the plasma cross section. The Green's function is written in terms of elliptic integrals [14],

$$G^\psi(R, Z, R_b, Z_b) = \frac{RR_b}{\pi\gamma} \frac{[(2-k^2)K(k^2) - 2E(k^2)]}{k^2}, \quad (4)$$

where $K(k^2)$ and $E(k^2)$ are the elliptic integrals of the first and second kinds, $\gamma^2 = (R + R_b)^2 + (Z_b - Z)^2$ and the argument is $k^2 = 4RR_b/\gamma^2$.

The contribution of the external coils can be written as,

$$\psi_c(R_b, Z_b) = \sum_{i=1}^{N_c} I_i G^\psi(R_i, Z_i, R_b, Z_b), \quad (5)$$

where N_c is the number of coils. In equation (5) we have assumed that all the coil currents (I_i) are concentrated at the points (R_i, Z_i) . It is also possible to consider that each coil has a certain cross section (S_i) and a current density $J_i = I_i/S_i$. Then the contribution of the coils to the poloidal flux can be calculated using equation (3), with the integral

extended over the coil cross section. The currents in the coils can be all different, or various coils can have the same current. This could be the case, for example, when the plasma equilibrium and the coil distribution are symmetric with respect to the $Z = 0$ plane (note, however, that the corresponding Green's functions will be different, except for $Z_b = 0$).

Now we introduce a function $\mathcal{E}$, defined as the square of the difference between the total flux calculated using equations (3) and (5) and the boundary value corresponding to the internal equilibrium,

$$\mathcal{E}(I_1, I_2, \dots, I_{N_c}) = \sum_{j=1}^{N_b} \left[\psi_p(R_b, Z_b) + \sum_{i=1}^{N_c} I_i G^\psi(R_i, Z_i, R_b, Z_b) - \psi_{in} \right]^2, \quad (6)$$

where the first sum is over all the boundary points considered (N_b) and ψ_{in} is the desired boundary value (typically taken to be zero). In equation (6) it is possible to introduce weight factors that control the importance assigned to each term. We have chosen to assign equal weight to all terms.

The basic idea is to determine the coil currents that minimize the value of $\mathcal{E}$, but there are some subtleties when the equilibrium has X-points. At the X-points, both derivatives of the poloidal flux, which are the components of the poloidal magnetic field, are zero. Then, small errors in the calculation of the boundary flux can produce large displacements of the X-points. One option [5] is to add to $\mathcal{E}$ a specific contribution coming from the value of the flux in the neighborhood of the X-point. The other is to specifically introduce the constraints related to the X-points in the minimization. This is the method employed, in a somewhat different context, in reference [15] and is the method we employ here. Specifically, we employ Lagrange multipliers to introduce the constraints $\psi(R_X, Z_X) = \psi_{in}$, $B_R(R_X, Z_X) = 0$ and $B_Z(R_X, Z_X) = 0$, where (R_X, Z_X) are the coordinates of the X-points.

To introduce the X-point constraints indicated above it is necessary to calculate both components of the poloidal field at the X-point. There are, again, two contributions, one due to the plasma and the other to the external coils. Fortunately, the Green's functions corresponding to B_R and B_Z contain the same elliptic integrals than the Green's function of the poloidal flux, with different coefficients. Reference [14] gives the equations for the components of the magnetic field produced by a current loop. In particular, equations (12)–(14) in that reference provide the cartesian components, which can be used to obtain B_R and B_Z . With our normalization the Green's functions become

$$G^{B_R}(R, Z, R_b, Z_b) = \frac{(Z - Z_b)}{2\pi R\gamma\alpha^2} \left[(R^2 + R_b^2 + (Z - Z_b)^2) E(k^2) - \alpha^2 K(k^2) \right], \quad (7)$$

$$G^{B_Z}(R, Z, R_b, Z_b) = \frac{1}{2\pi\gamma\alpha^2} \left[(R_b^2 - R^2 - (Z - Z_b)^2) E(k^2) + \alpha^2 K(k^2) \right], \quad (8)$$

where $\alpha^2 = R_b^2 + R^2 + (Z_b - Z)^2 - 2RR_b$ and G^{B_R} and G^{B_Z} are the Green's functions for B_R and B_Z respectively. With these Green's functions the contribution of the plasma to B_R and B_Z is obtained using equation (3), with G^ψ replaced by G^{B_R} or G^{B_Z} and the contribution of the coils using equation (5).

The function employed in the constrained minimization is defined as,

$$\mathcal{L} = \mathcal{E} + \lambda_1 B_R(R_X, Z_X) + \lambda_2 B_Z(R_X, Z_X) + \lambda_3 [\psi(R_X, Z_X) - \psi_{in}], \quad (9)$$

where λ_1 , λ_2 and λ_3 are the Lagrange multipliers. Deriving $\mathcal{L}$ with respect to the coil currents we obtain N_c equations,

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial I_l} = & 2 \sum_{j=1}^{N_b} \left[\sum_{i=1}^{N_c} I_i G_i^\psi(R_{b_j}, Z_{b_j}) + \psi_p(R_{b_j}, Z_{b_j}) - \psi_{in} \right] G_l^\psi(R_{b_j}, Z_{b_j}) \\ & + \lambda_1 G_l^{B_R}(R_X, Z_X) + \lambda_2 G_l^{B_Z}(R_X, Z_X) + \lambda_3 G_l^\psi(R_X, Z_X) = 0, \\ & l = 1, \dots, N_c. \end{aligned} \quad (10)$$

Deriving $\mathcal{L}$ with respect to the three Lagrange multipliers we obtain three additional equations,

$$\frac{\partial \mathcal{L}}{\partial \lambda_1} = B_R(R_X, Z_X) = 0, \quad (11)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_2} = B_Z(R_X, Z_X) = 0, \quad (12)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_3} = \psi(R_X, Z_X) - \psi_{in} = 0. \quad (13)$$

This gives a total of $N_c + 3$ linear equations to calculate the coil currents and the Lagrange multipliers. Since the number of coil currents is usually small ($N_c < 20$) it is very easy to solve this system of equations.

Once the coil currents are known it is very simple to calculate the poloidal flux and field in the outer region using equations (3) and (5) and changing (R_{b_j}, Z_{b_j}) by the coordinates of the outer points.

3. Example

To show how the method works we present an example of the calculation of the coil currents and external poloidal flux and field for an equilibrium with parameters and external coil distribution similar to those of the DIII-D tokamak [16]. The plasma (internal) equilibrium was obtained using the method described in [13]. It is a static (no rotation) equilibrium which is symmetric with respect to the $Z = 0$ plane and has two X-points. The parameters characterizing this equilibrium are: major radius $R_0 = 1.65$ m, minor radius $a = 0.60$ m, vacuum toroidal field at R_0 $B_0(R_0) = 2.0$ T, elongation $\kappa = 1.98$, triangularity $\delta = 0.55$, squareness $\zeta = 0.3$, plasma current $I_p = 1.5$ MA, and $\beta = 0.04$. The corresponding normalized values of the relevant geometrical parameters

are: $R_i = 0.6364$ (inner equatorial point), $R_o = 1.3636$ (outer equatorial point), $R_X = 0.8$ and $Z_X = \pm 0.72$. The positions and calculated currents of the external coils located in the upper half-plane are given in table 1. The fourth column shows the normalized currents and the fifth one the values in MA. Note that the coils can have many turns so the actual current flowing in the conductors can be significantly smaller. The coordinates correspond approximately to the center of the field shaping coils shown in figure 1 of [17] (the central solenoid is not included). We assume that there is another set of 9 coils with the same currents located symmetrically with respect to the $Z = 0$ plane.

The contribution of the coils to the poloidal flux and field is calculated using equation (5), with the corresponding Green's functions for B_R and B_Z . The only change is that the contribution from each pair of symmetric coils is grouped together. For example, if we assume that coils #1 and 10 are symmetric with respect to the $Z = 0$ plane and have the same current, their contribution to the poloidal flux will be $I_1 [G^\psi(R_1, Z_1, R_b, Z_b) + G^\psi(R_{10}, Z_{10}, R_b, Z_b)]$. Thus, equations (10)–(13) give us a total of 12 equations to calculate the 9 unknown currents and the three Lagrange multipliers.

Table 1. Positions and currents of the 9 coils located in the half plane $Z > 0$.

Coil	R	Z	I	I (MA)
1	0.6080	0.9193	2.50448×10^{-1}	0.6580
2	0.7588	0.9711	-1.138809×10^{-2}	-0.0299
3	1.0236	0.9591	-4.758426×10^{-2}	-0.1250
4	1.5020	0.6770	-1.37177×10^{-1}	-0.3604
5	1.5833	0.2652	-7.754115×10^{-2}	-0.2037
6	0.5217	0.1030	-4.753619×10^{-2}	-0.1249
7	0.5217	0.3091	-6.576987×10^{-2}	-0.1728
8	0.5217	0.5152	-8.124016×10^{-2}	-0.2134
9	0.5217	0.7212	-1.76024×10^{-1}	-0.4624

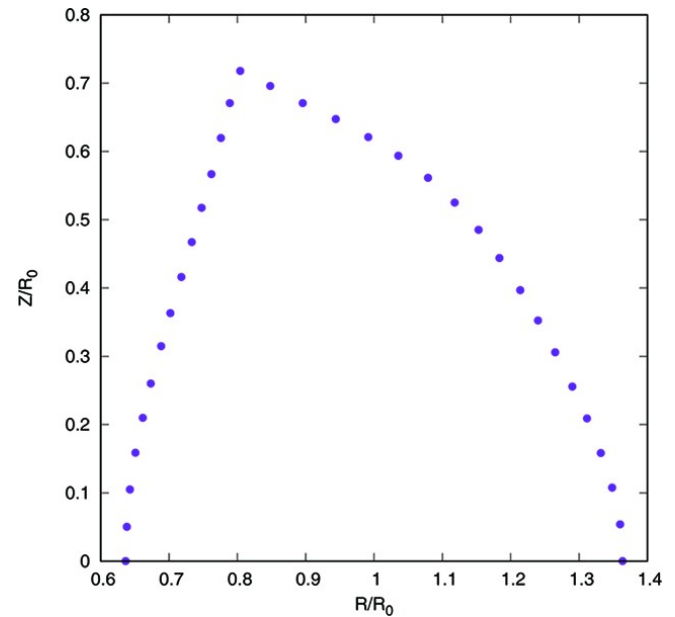


Figure 1. Points along the plasma boundary employed to calculate the error.

Figure 1 shows the points along the plasma boundary used to calculate $\mathcal{E}$. The X-point is also included to obtain the correct shape for the plasma boundary but it is not employed in the calculation of $\mathcal{E}$ because its position is fixed with the Lagrange multipliers. A total of 33 points along the plasma boundary were employed in this example, including both equatorial points. Introducing a relative error e_r , defined as

$$e_r = \frac{1}{N_b \psi_{\max}} \left\{ \sum_{j=1}^{N_b} \psi^2(R_{b_j}, Z_{b_j}) \right\}^{1/2}, \quad (14)$$

where $\psi_{\max}$ is the flux at the magnetic axis we obtained $e_r \approx 2 \times 10^{-3}$ with $N_b = 33$ and using a grid of 100×100 to calculate the contribution of the plasma current.

Figure 2 shows the poloidal flux contours in the upper half plane ($Z > 0$), both inside and outside the plasma. The X-point is at the exact prescribed position. The positions of coils #1, 3 and 6–9 (see table 1) can be inferred from the closed flux surfaces formed around them. Coil #2 has a very small current and produces only a very small disturbance in the flux surfaces. We note that although a relatively large external region is shown, the vacuum vessel wall is usually fairly close to the plasma boundary. Finally, figure 3 shows $B_R(R = R_X, Z)$ (red) and $B_Z(R = R_X, Z)$ (blue). It is clear that both components of the poloidal field are continuous and become zero at the X-point, as requested.

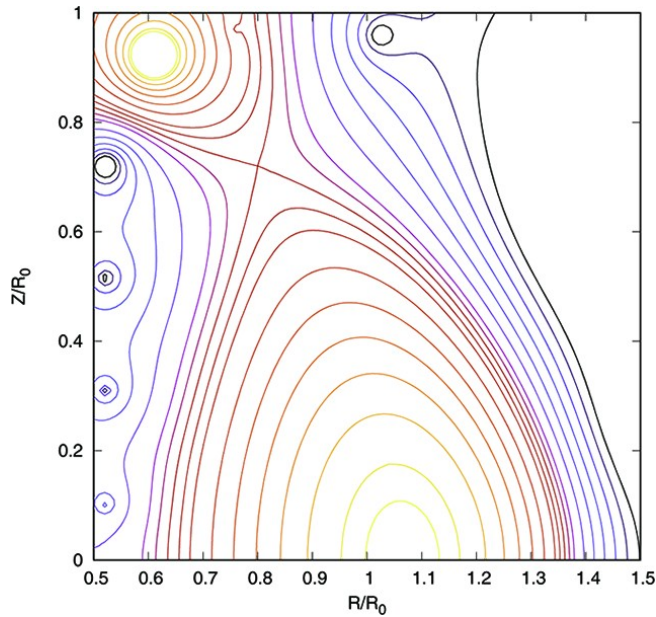


Figure 2. Poloidal flux contours in the upper half plane. The X-point and the location of some external coils are also shown.

4. Summary

A simple and fast method to calculate the currents that have to flow in a set of external coils to obtain an external magnetic configuration consistent with a prescribed internal equilibrium was presented. The method involves a

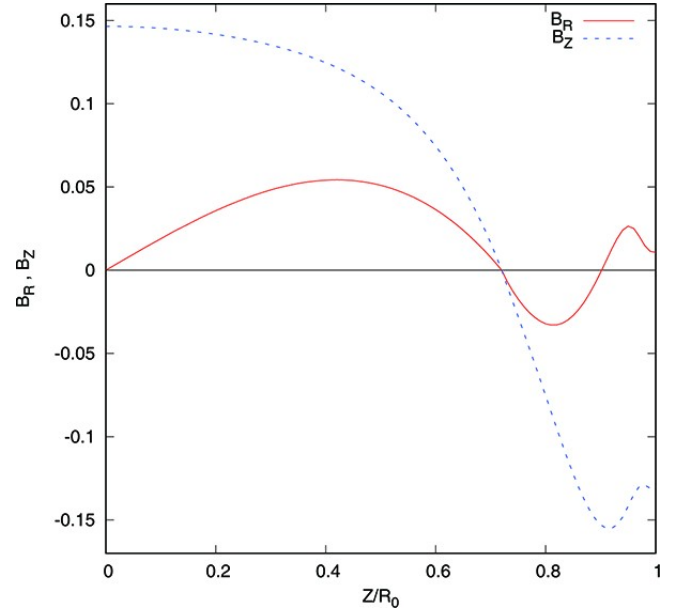


Figure 3. B_R (red) and B_Z (blue) plotted as a function of Z for $R = R_X$. Both components become zero at the X-point ($R = 0.80$, $Z = 0.72$).

constrained optimization and does not require the use of finite elements or sophisticated numerical techniques. The poloidal flux and field are calculated using Green's functions expressed in terms of elliptic integrals that are calculated using a fast and accurate routine. The calculation can be restricted to the region between the plasma and the device wall, it is not necessary to extend it to a large region to satisfy the boundary conditions at infinity. The example presented shows that the constraints on the boundary conditions and, in particular, on the position of the X-point, are accurately satisfied. A possible application of the method presented here is the rapid calculation of different divertor configurations.

Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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